

School of Mathematics and Statistics, UNSW
MATH3311/MATH5335, Python

Exercises 1: Computing with real number

1. Floating-point number systems are only finite approximations to the real number system, based on a finite binary representation of numbers. Thus, powers of 2 are represented exactly, but powers of 10, for example, 0.01, might not have a finite binary representation and so must be approximated by rounding to the closest floating-point number.

Look at the IPython console in the lower right of the Spyder window. Click on the 'hamburger button' (icon consisting of three horizontal lines) and from the drop-down menu select 'Special consoles' followed by 'New PyLab console (data plotting)'. You should see a message 'Populating the interactive namespace from numpy and matplotlib'.

Try the following in Spyder's IPython console.

- (a) `-4**2, 0 - 4**2`
 - (b) What is the difference between `-1 * A` and `A` if `A = array([1, -2, 3])`?
 - (c) A student calculates $x = \frac{a}{R+r}$, where $a = 2$, $R = 4$ and $r = 2$, and finds that $x = 2.5$. Is this answer correct?
 - (d) `0.75 * 0.2 - 0.15`
 - (e) `floor(6/1)` and `floor(0.6/0.1)` where `floor(x)` is the largest integer less than or equal to x .
 - (f) $x = 100$, $h = 1e - 14$, $x + h - x$, $x - x + h$
 - (g) $x = 100$, $h = 1e - 15$, $x + h - x$, $x - x + h$
 - (h) `print(finfo(float))`
 - (i) `eps = finfo(float).eps, 1 + eps > 1, 1 + eps/2 > 1`.
2. Consider the equation $\pi^7 = e^x$.
 - (a) Solve the equation by hand to get a mathematical expression for x .
 - (b) Use the Python to evaluate this expression for x . (`from numpy import log, pi`)
 - (c) Test if π^{**7} and e^x are equal using the relational operator `==`.
 - (d) Evaluate $|\pi^7 - e^x|$. (`from numpy import exp`)
 - (e) Estimate the rounding error in storing π^7 on a computer using
 - IEEE single precision arithmetic, (`from numpy import finfo, float32`)
 - IEEE double precision arithmetic. (`from numpy import float64`)
 3. IEEE floating-point arithmetic has a special representation for nan ('Not-a-Number'), which is used as the value for quantities that are not well-defined mathematically. There are also special representations `inf` for ∞ . If an operation produces one of these values then they can propagate throughout subsequent calculations.

Try the following in Spyder's IPython (Pylab) console.

- (a) `1/0, -1/0, log(0), log(-1)`
- (b) `z = float64(0), 1/z, -1/z, log(z)`
- (c) `0/0, z/z, inf - inf, 0 * inf, 0 * nan, exp(-inf)`

4. Let $f(x) = \sin(\pi x)$.

- (a) What are the mathematical values of $f(k)$ for $k = 0, 1, \dots, 100$?
- (b) Use Python to calculate $f(k)$ for $k = 0, 1, \dots, 100$. Did you expect this result? (`from numpy import arange, sin, pi`)

5. Find the largest double precision positive values of x and t such that

- (a) e^t is finite,
- (b) $e^{-t} > 0$,
- (c) e^{x^2} is finite,
- (d) $e^{-x^2} > 0$. (`from numpy import finfo, log, sqrt`)

6. The integral

$$J_n = \int_0^1 x^n e^{x-1} dx$$

can be found for successive integer values of n using the recursion formula

$$J_n = 1 - nJ_{n-1} \quad \text{for } n \geq 1, \text{ with } J_0 = 1 - e^{-1},$$

established via integration by parts. Calculate J_n for $n = 1, 2, \dots, 10$, using

- (a) the approximate value $e^{-1} = 0.367879$, (`from numpy import empty, exp`)
- (b) the value given by `exp(-1)`.

This is an example of a calculation that is sensitive to error in the initial data. There are other calculations, such as evaluating e^{-x} for large positive x from the Taylor series

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots,$$

that are sensitive to roundoff error, although the double precision arithmetic used in many packages can hide much of this sensitivity.

7. When using Spyder, you can obtain help about a function by

- typing `?` followed by the name of the function at the IPython prompt, or
- typing the name of the function followed by `.`

Find out information about the commands used above, including

- (a) `finfo`
- (b) `floor`, `ceil`
- (c) `exp`, `log`, `log10`, `log2`, `log1p`

8. The fastest available commodity microprocessors have CPUs with a sustained max clock speed in the range 2-4 GHz, where 1 Hz is a clock cycle per second, and 1 GHz is cycles/second. Physical limitations, including heat production, limit how much faster CPUs can be clocked, so modern CPUs have several cores that can operate in parallel. Additionally, the arithmetic units may support special instructions enabling one or more floating-point operations per core per clock cycle.

Suppose you have a 2.5GHz quad core CPU where each core can do 2 floating point operations per clock cycle.

- (a) How long is a single clock cycle in nanoseconds (10^{-9} seconds)?
- (b) The speed of light in a vacuum is $c = 2.99792 \times 10^{10}$ cm/sec. Estimate how far an electronic signal in a circuit can travel in 1 ns.
- (c) What is the size n of the largest matrices $A, B \in \mathbb{R}^{n \times n}$, that can be multiplied (using $2n^3$ operations) in each of the following time intervals?

- 1 second
- 1 minute
- 1 hour
- 1 day

- (d) What is the largest number of cities n in a Travelling Salesman Problem (TSP) that can be solved by simple enumeration requiring $n!$ operations in each of the following time intervals?

- 1 second
- 1 minute
- 1 hour

9. A double precision floating point number takes up 8 bytes (64 bits) of RAM. What is the size n of largest $n \times n$ matrix that can be stored in

- (a) 2GB of RAM?
- (b) 4MB of Level 2 cache memory?
- (c) 32KB of level 1 cache memory?

10. A 32 bit operating system uses 32 bit pointers to address item stored in RAM, whereas a 64 bit operating system uses 64 bit pointers.

What is the largest amount of memory (RAM) that can be directly addressed on a

- (a) 32 bit operating system?
- (b) 64 bit operating system?